Desk controller

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Change history

1.0.0	23/09/2026	First release
1.1.0	18/10/2026	Migration to FLRHW KiCad starter template

System overview



File: System_overview.kicad_sch

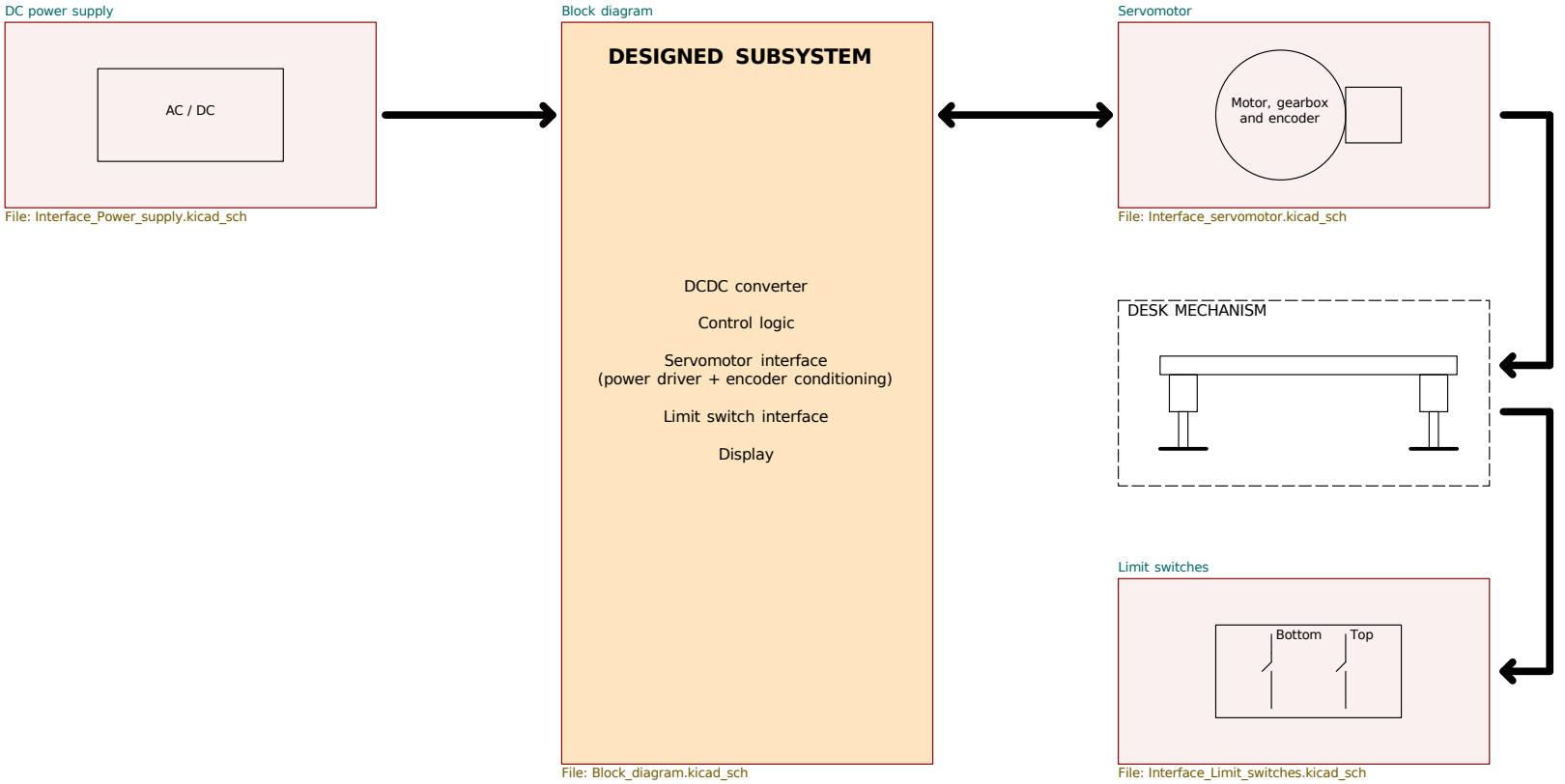
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Desk controller

Notes

Sheet: Root	
File: Desk_controller.kicad_sch	
Rev: 1.1.0	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-18	Rls'd: FLR
Dwg: NA	Size: A4
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System overview



This sheet shows the overall system and the interfaces to external components.
External components are shown for system context only and are not part of this schematic.
Only the "DESIGNED SUBSYSTEM" is covered by this schematic.

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Desk controller

Notes

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Rev: 1.1.0	Drwn: FLR
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DC power supply

Interface specification

Barrel connector (Dell type, with ID pin)

Pin	
1	Supply - 19.5 V, 3 A (inner shell)
2	ID (internal pin)
3	GND (outer shell)
SH	Shield

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Notes

Sheet: DC power supply	
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Dwg: NA	Size: A4
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Servomotor

Interface specification

JST B6B-XH-A / JST XHP-6 connector

Pin	Description
1	Encoder B output (VCC)
2	Encoder A output (VCC)
3	GND
4	VCC (3.3 - 5V)
5	Motor B (3 A, 12-24 V)
6	Motor A (3 A, 12-24 V)

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Desk controller

Notes

Sheet: Servomotor	
File: Interface_servomotor.kicad_sch	
Rev: 1.1.0	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-18	Rls'd: FLR
Dwg: NA	Size: A4
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Limit switches

Interface specification

JST B4B XH A / JST XHP-4 connector

Pin	Description
1	Bottom limit switch (Normally open, switch closes to GND)
2	GND
3	GND
4	Top limit switch (Normally open, switch closes to GND)

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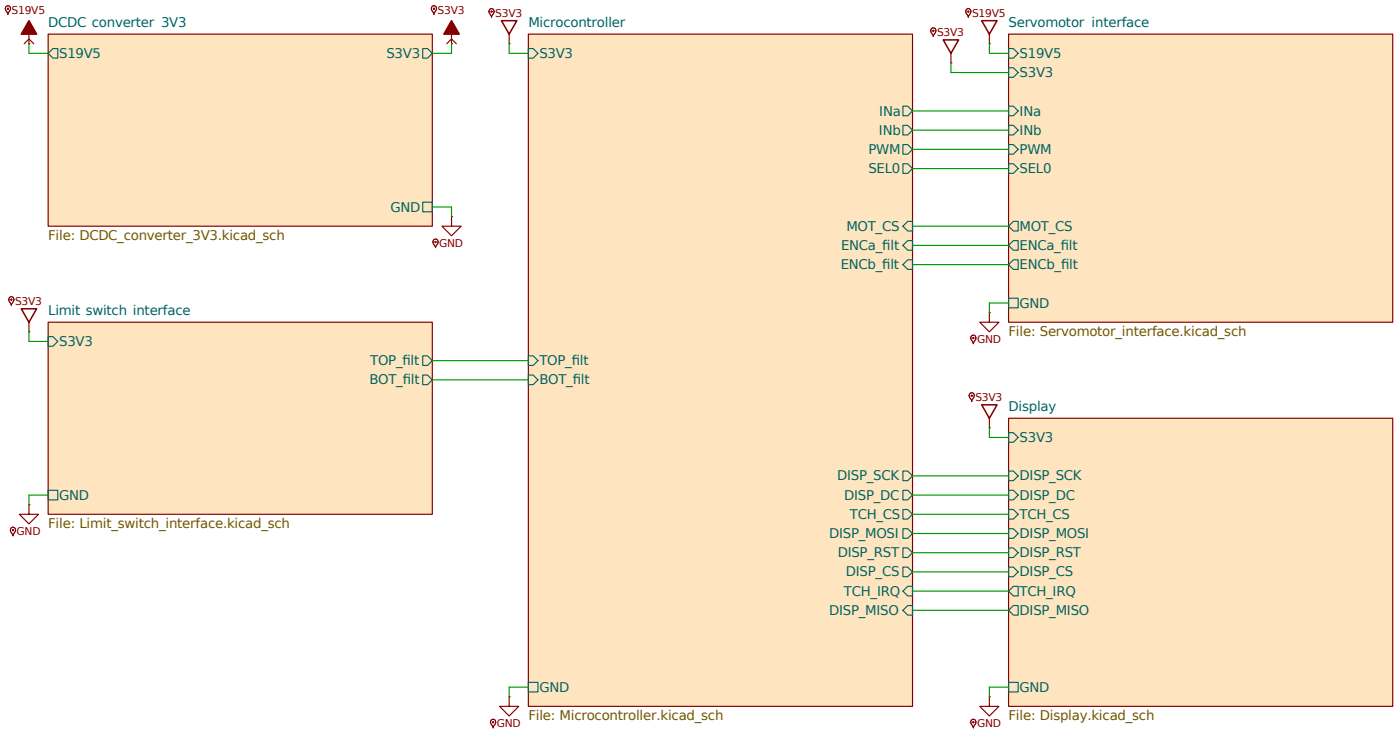
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Desk controller

Notes

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Block diagram



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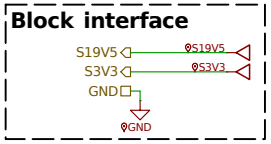
Desk controller

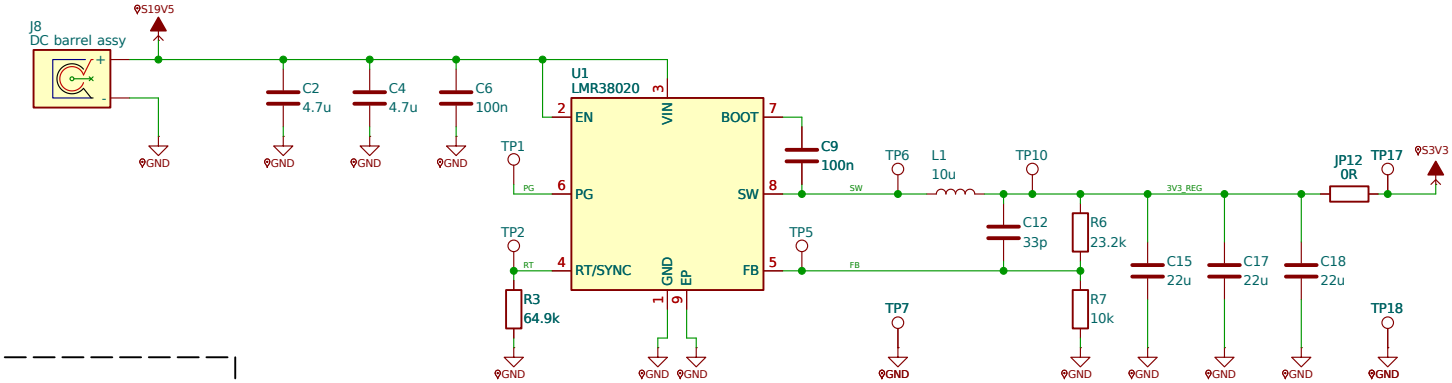
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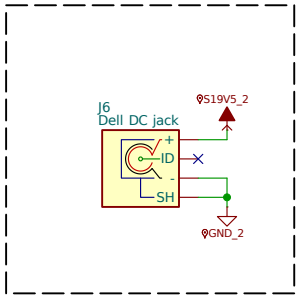
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Mat: Concept	Rev'd: FLR
Date: 2026-10-18	Rls'd: FLR
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DCDC converter 3V3

Block interface







J6 is fitted to a small "standalone" PCB which will then be soldered into the "main" PCB.
This standalone PCB + connector are what make the "DC barrel assy" used in J8 position.

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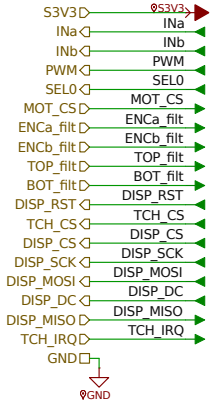
Desk controller

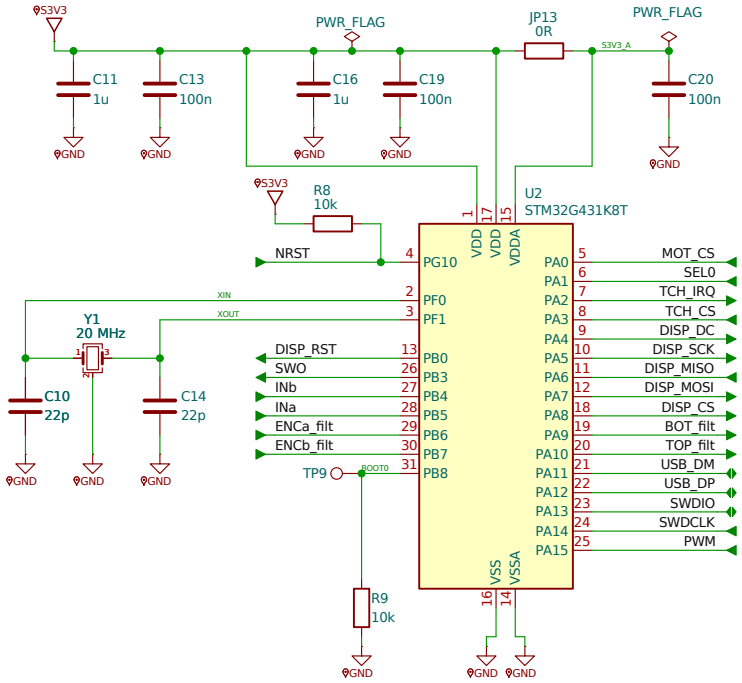
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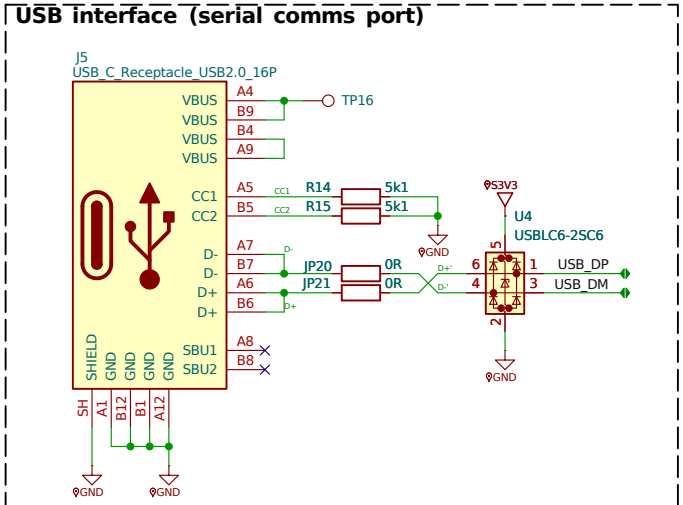
Microcontroller

Block interface

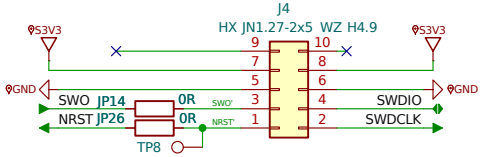




USB interface (serial comms port)



SWD header



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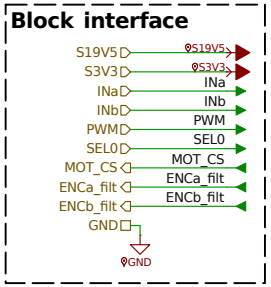
Desk controller

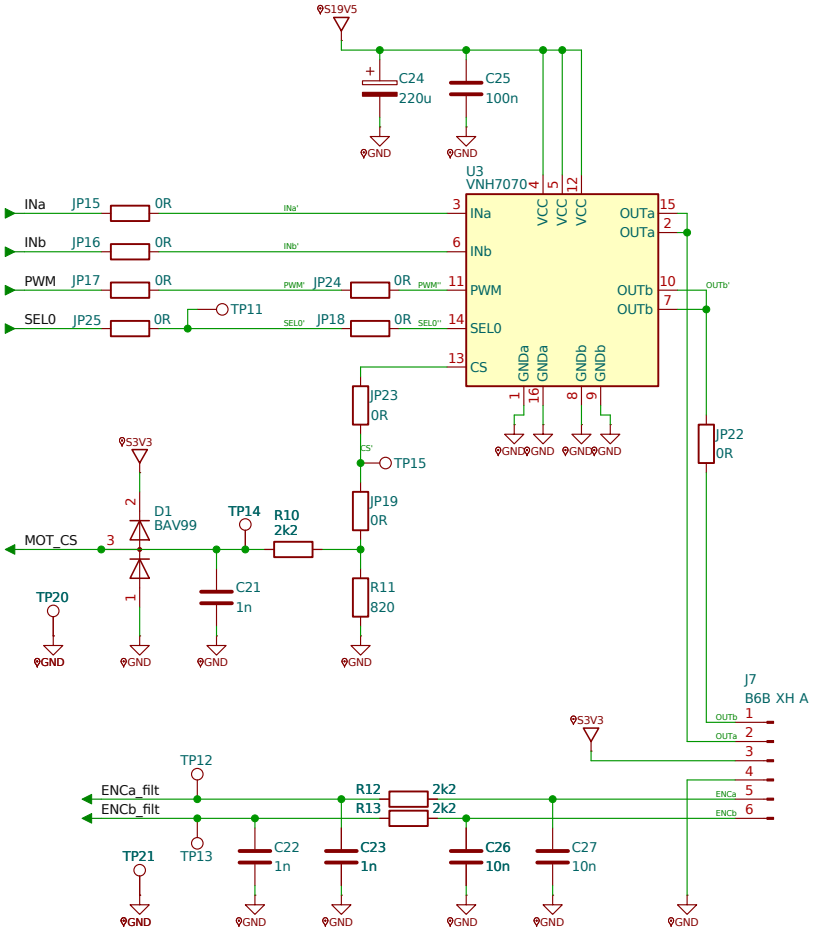
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Servomotor interface

Block interface



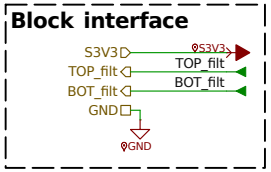


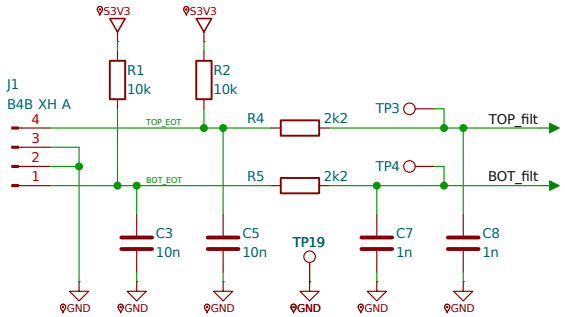
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Limit switch interface

Block interface





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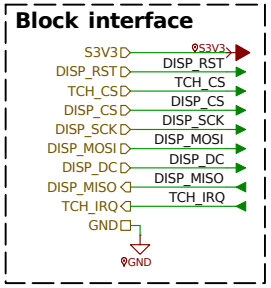
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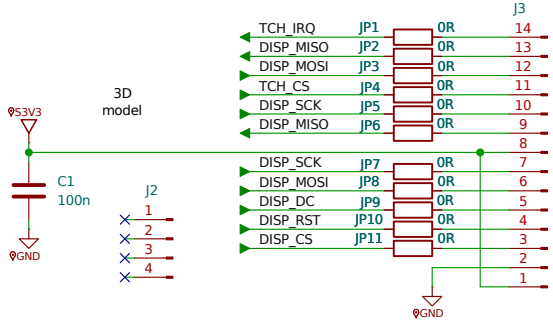
Desk controller

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Display





TFT1
TFT_2.4in_320x240

The display will not be directly soldered to the PCB. Instead, generic 2.54 mm pitch sockets J2 and J3 will be used as connectors.

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Desk controller

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